

Ornithology from the Lakeshore

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Ornithology from the Lakeshore

PUBLISHING ANECDOTAL INFORMATION

The plural of anecdote is data

Raymond Wolfinger (graduate course Stanford University, 1969-1970)

The plural of 'anecdote' is not 'data'

Irwin S. Bernstein (1988)

On 30 December 2019, the prestigious, multidisciplinary journal 'Proceedings of the National Academy of Sciences of the United States of America', or PNAS, published a brief paper entitled 'Evidence of Tool Use in a Seabird' (Fayet *et al.* 2019). The paper reports on two independent observations of an Atlantic Puffin *Fratercula arctica* using a wooden stick to scratch its body. The first observation took place at dusk on 18 June 2014 near a breeding colony on Skomer Island in Wales. Using a spotting scope, researchers observed a puffin floating in the sea beneath the cliff where the colony was located. The bird held a stick in its beak and used it to scratch its back for about five seconds before flying off with the stick still in its beak. A motion-activated camera captured the second observation at a breeding colony on Grimsey Island, the northernmost inhabited part of Iceland. In a video recorded on 13 July 2018, an adult puffin can be seen picking up a stick from the ground, walking with the stick in its beak, bending its head downward and then briefly touching its belly feathers with the stick. These descriptions of the behaviour of the two individual puffins comprise the entire results section of the published article. The paper can therefore be confidently called an anecdote, meaning "an account of a rare behaviour or event that has been observed either once or a few times" (Sándor & Miklósi 2020, see also Sarrinhaus *et al.* 2005).

Anecdotal reports are widespread, not only in the fields of animal behaviour and behavioural ecology, but also in other areas of research, including medicine, but their value has often been questioned. As Merriam-Webster (www.merriam-webster.com/dictionary/anecdotal%20evidence) puts it, "His conclusions are not supported by data; they are based only on *anecdotal evidence*". Anecdotes have a bad reputation in the behavioural sciences. They are often dismissed as storytelling, and, indeed, sometimes an anecdote is merely a story that can be placed in the same category as a myth or folklore. Because we like a good story, anecdotes can be paradoxically persuasive. The claim that Australian

raptors actively spread fires by carrying a piece of burning wood in their talons and dropping it elsewhere to start a new fire is a case in point (see Kempnaers 2022). Although even such stories can have their value if they stimulate us to look more closely at a phenomenon, a scientific anecdote is something quite different: it is a detailed description of an observation (Bates & Byrne 1988), often backed up with photo or video material, such as in the report on the Atlantic Puffin.

Anecdotal records are also criticized because they do not prove anything, with the argument that rigorous scientific evidence requires systematic, larger-scale data, with a statistical test of a hypothesis. Furthermore, anecdotes are often considered to be biased, in the sense that they are selectively reporting an "outlier", a rare behaviour. Sándor & Miklósi (2020) go as far as to suggest that an anecdote may be nothing more than an "anomaly", and that instead of anecdote we should use the term "behavioural rarity". Although this might be appropriate in some cases, the suggestion does not do justice to most scientific anecdotes.

The negative connotation attached to an anecdotal report and the declining number of publications of such anecdotes (at least in primatology; Ramsay & Teichroeb 2019) is unfortunate, because anecdotal information can have a major scientific impact. First, an anecdotal report can serve as an important source of discovery. It can be the spark for further work, the starting point to generate new, testable hypotheses, leading to novel, and possibly groundbreaking research. Many breakthroughs in medical research, such as the discovery of insulin, began as scattered, informal observations (Parodi & Parodi 2014).

Second, a single anecdote can serve as a 'black swan', falsifying what was thought to be generally true. For example, it was widely believed that birds lack the cognitive capacity to build tools, until a doctoral student at the Department of Ecology of Massey University in New Zealand discovered that free-living New Caledonian Crows *Corvus moneduloides* did exactly that (Hunt 1996).

Third, an anecdote can demonstrate what is possible, even if the behaviour is uncommon. For example, researchers from the Coastal Carolina University and the South Carolina Department of Natural Resources documented six cases in which an unpaired, male Loggerhead Shrike *Lanius ludovicianus* adopted a brood of nestlings or fledglings. In two cases, the male later became the social partner of the mother of those offspring, suggesting that this form of alloparental care might be adaptive (Hill *et al.* 2023). Other examples are anecdotal reports showing that some male Blue Tits *Cyanistes caeruleus* care for offspring at a nest where they fathered extra-pair offspring (Kempnaers 1993, Santema & Kempnaers 2021). The circumstances under which these behaviours occur may well be rare, but the observations provide valuable insights into the mechanisms individuals use to make decisions, and about the potential fitness benefits of alloparental care and of female extra-pair behaviour. An anecdotal report may also describe behaviour that is rarely observed, but in fact quite common, such as ‘anting’, infanticide, roosting in particular places, predation at night or high in the air (e.g. Bom *et al.* 2023, Boom *et al.* 2024). Many interesting behaviours are hard to detect, difficult to observe, brief, or occur during periods when we are rarely watching (e.g. at night, or during a particular part of the year).

Fourth, an anecdote may be the first description of a behaviour that just arose but is spreading in a population. A classic example is the opening of milk bottles by tits, a behaviour that spread through the UK in the early 20th century (Fisher & Hinde 1949). Another example is the peculiar egg-laying behaviour of the endangered Black Robin *Petroica traversi*: a female put its eggs on the rim instead of in the cup of the nest. Researchers that checked robin nests moved such eggs inside the cup, resulting in the behaviour becoming more frequent (Massaro *et al.* 2013).

Finally, only considering an anecdotal observation in isolation does not do justice to the fact that collections of such reports can build robust datasets that can be used to describe behaviour, breeding biology and diet (e.g. Birrer 2009) and to formally test hypotheses. For example, one can collect all anecdotes about tool use in birds, analyse how taxonomically widespread it is, and test which ecological factors those species that show the behaviour have in common. Anecdotes can thus be considered data that form the foundation for further scientific research.

Here, from the lakeside, let me make a strong plea for publishing ornithological anecdotes, even if it only puts you on the lower rungs of the academic ladder of

esteem. Anecdotal reports can have great value, but – as with all scientific papers – they vary widely in quality. So, let me highlight and discuss the key criteria to consider when writing, reviewing or reading a report of anecdotal information. Following a simple recipe with three main ingredients is all it needs to make an excellent anecdote.

First, ensure that the behaviour is described precisely, together with the circumstances under which it occurred, and add all relevant details about the methods. Ideally, include photographic or video material, because this allows independent evaluation of the description and interpretation of the observation. Or, as Frans de Waal put it, “I have nothing against anecdotes, especially if they have been caught on camera or come from reputable observers who know their animals” (de Waal 2016). By the way, these “reputable observers” can also be amateur ornithologists, many of whom publish valuable anecdotal information in local bird journals.

Good examples of precise description can be found in the two anecdotal reports in this issue of Ardea. The first one describes a case in which a Blue Tit nestling is raised by a Eurasian Nuthatch *Sitta europaea* pair together with five nuthatch nestlings (Velasco & Sanz 2026). Velasco & Sanz checked the nestbox regularly from the early nest building stage all the way until fledging. They observed that two Blue Tit eggs appeared simultaneously with the first two Nuthatch eggs. Later, they found the female Nuthatch incubating a clutch consisting of six Nuthatch and two Blue Tit eggs. The authors measured and weighed the nestlings and show that the Blue Tit raised by the Nuthatch pair was underdeveloped in size (tarsus and wing length), but similar in body mass, compared to Blue Tit nestlings of similar age found in the same population and year, but raised by their own parents. The second anecdotal report describes an unusual case in which a female Japanese Tit *Parus cinereus minor* incubates an empty nest for at least 20 days (Park *et al.* 2026). The authors describe that the female started sitting in the empty nest cup on 14 June 2019, two days after nest building started, and continued what looked like incubation behaviour until 3 July. The authors describe how often they visited the nest, when they filmed inside and outside of the nestbox, and what they observed. They also give background information on the species and population, such as the timing of the breeding season, the variation in clutch size, and the typical duration of the incubation period.

Second, make sure that the precise description of the behaviour is clearly separated from the interpreta-

tion of that behaviour. For example, Park *et al.* (2026) do not simply state that the female showed “incubation behaviour”, but they define what they mean by this term and describe what they actually observed (e.g. vocal interactions between the male and the female, the female entering the nestbox and the male following her) and what they did not (e.g. the male feeding the female). In the story about the Atlantic Puffins, picking up a stick and moving it such that it touches the feathers is an observation. Writing that the bird used the stick to scratch its body already feels like an interpretation.

Third, invest some thought into putting the observations in context. The value of an anecdotal report increases when the relevance of the observation is clarified, and when tentative, but testable hypotheses can be formulated. For example, Park *et al.* (2026) discuss potential mechanisms related to the onset and end of incubation. They highlight that the incubation behaviour persisted for 20 days and that this is longer than the usual 12–15 days. This may be true, but another unusual aspect of this case is that the female already started incubation two days after the start of nest building. I assume that nest building typically lasts longer, but even if not, female Japanese Tits lay at least five eggs (and up to 12), which implies that the female still ended incubation before or (the latest) at the expected hatch date. The authors also propose some ideas for further research about factors that may influence hormone levels that underlie the behaviour. Furthermore, Park *et al.* (2026) present indirect evidence that the male fed the female during her incubation period, and they propose that females could simulate incubation to exploit the male to provide food. Although this idea seems rather farfetched, it does provide food for thought about the cues males use to invest in provisioning their mate.

In the other anecdotal report, Velasco & Sanz (2026) discuss several potential explanations for the slower development of the Blue Tit nestling in the mixed brood, related to diet and to competition among the nestlings. They also highlight that the Nuthatches did not discriminate between their own and the Blue Tit eggs and nestlings. Finally, they raise intriguing questions about mixed broods, in particular about the future reproductive potential of the foreign nestling(s). For example, how will the vocalizations and song develop in a Blue Tit raised by a Nuthatch pair?

Although it is important to put the observations in a wider context, the possibility of publication in a higher-impact journal may tempt some authors to draw far-reaching conclusions. However, overinterpreting your

observations will only add bad-reputation fuel to the story-telling fire. The publication of the puffin paper caused some consternation among scientists, mainly because of how the authors interpreted their observations. Fayet *et al.* (2019) highlight that their observation is only the second example of “body-care-related tool use” in a wild bird and conclude that “the wild avian tool-use repertoire is wider than previously thought”. They point out that the “independent observations span a distance of more than 1,700 km” and conclude that “occasional tool use may be widespread in this group and that “seabird’s physical cognition may have been underestimated.” Several papers were published in response to the original article. Among them were critical commentaries that questioned the conclusions, mainly because of disagreement about what the bird in the video was doing and what kind of cognitive skills were involved (Auersperg *et al.* 2020), but also pointing to an earlier report on tool use in another free-living seabird (Meyerrieks 1972). The publication by Fayet *et al.* (2019) stimulated even more work: it led to a paper that used the controversy to provide guidelines on how to report anecdotal observations (Sándor & Miklósi 2020), and to another paper that used the example to study whether experts (professional ornithologists or ethologists) evaluate an observation (the behaviour seen in the video) more objectively, accurately and coherently than lay people (Sándor *et al.* 2021).

In conclusion, when evaluating the merits of a particular anecdotal report, be reminded about the origin of the word anecdote: it comes from the Greek ‘anekdota’, meaning ‘unpublished things’. Indeed, not all anecdotes are suitable for publication.

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